

BO3:

Workload-Specific Compilation

Pipelines for Multi-Level Compilers

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All 9 workloads beat IREE -03 · 5.66× geomean

1. One pass, moved one stage earlier, ran 2.4× faster

ResNet-50 (one of three models probed this way), one pinned x86 core, knobs fixed. We insert linalg-generalize-named-ops as a probe pass and change only where in the pipeline it runs.

Where the probe pass runs	Compute density (FMA%)	Latency
not inserted (baseline)	52.7	1595 ms
early — at preprocessing	45.6	662 ms
late — at dispatch creation	77.8	1575 ms

The lowest compute density won by 2.4×. FMA% does not predict the winner.

2. Each model’s best pipeline was a different pipeline

	resnet50	mobilenet v2	yolo v5	frCNN	deeplab 513x513	bert base 128
T conv+pad+no tile	2.84×	1.41×	1.28×	2.27×	3.68×	0.99×
I+no tile	1.95×	2.33×	8.72×	17.21×	0.99×	1.04×
CC+I+mask+no tile	2.15×	1.43×	9.08×	9.44×	2.65×	1.00×
I+peel+tile	0.98×	1.54×	7.86×	19.91×	FAIL	1.12×
T conv+mask+no tile	2.65×	1.38×	1.23×	2.94×	3.89×	1.00×
gen+tile+unroll	2.14×	2.23×	7.75×	19.20×	FAIL	1.36×

Motivating study: 6 fixed pipelines × 6 models. Each cell is that pipeline’s speedup over -03 on that model. Blue marks the model’s fastest pipeline; all six differ. Red marks its slowest.

frCNN’s best pipeline makes resnet50 slower than -03 (0.98×) and fails on deeplab.

Pass bundles: gen = generalize, CC = convert-conv-to-channels-last, T = transpose-convolution, I = convert-conv2d-to-img2col.

3. BO3 searches one pipeline per workload, in three stages

I. SCREEN

Benchmark the 4 backend knobs: unroll, tile, vector pre-proc, vectorization. Keep the top 3; each is an anchor.

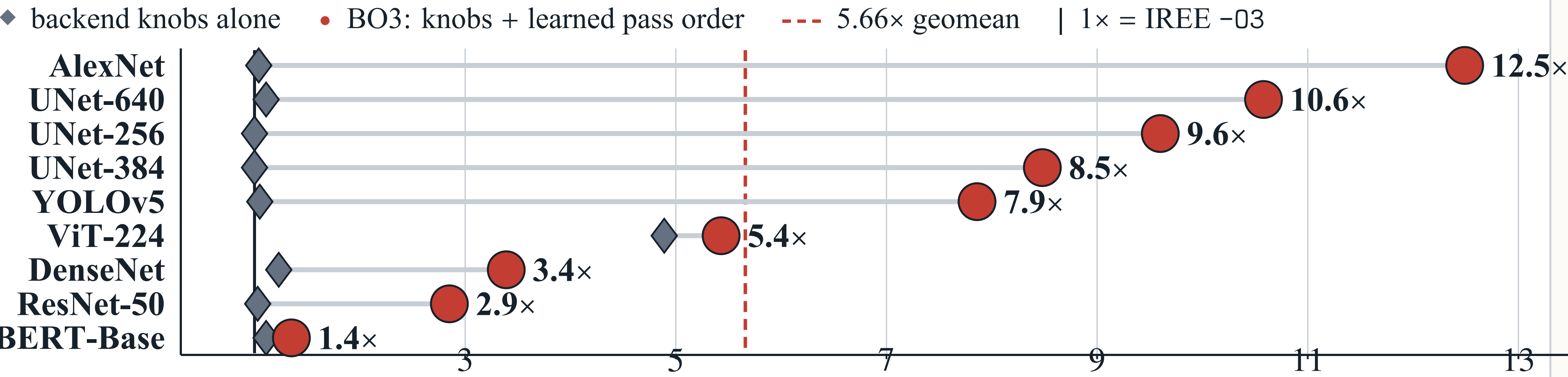
II. LEARN

A GNN policy reads the IR’s def-use graph and picks the next legal pass from a 9-pass vocabulary. Reward is measured latency.

III. SELECT

Sample 10 rollouts per anchor. Report the fastest candidate across all 3 anchors.

4. All nine workloads beat -03, by 1.4× to 12.5×



Speedup over -03, Top-1 anchor. The gap in each row is what the learned pass order adds on top of the knob anchor; it is the larger of the two contributions on 8 of the 9. ViT-224 is the exception, where knobs alone already reach 4.89×. Matched 610-rollout budget on UNet-640: random legal search 14.8 s, BO3 1.71 s.

TAKEAWAY

The best pass pipeline is specific to the workload.

BO3 searches one per workload, scored on measured latency.

Offline per-workload tuner · one pinned x86 core · IREE 3.9-rc · Stage II training 5.83 h geomean per knob-conditioned task.

